

Table VS3. Structural overlap and nearest-neighbour similarity between the sixty designed compounds and the training set.

The training set contains 1,811 unique molecules with 618 distinct ring-containing Bemis–Murcko scaffolds. Exact matches were counted after canonicalising the training and screening SMILES with RDKit 2026.03.4. Scaffold matches were counted using Bemis–Murcko scaffold SMILES. For each screening compound, the nearest-training-molecule similarity is the largest Tanimoto coefficient against the 1,811 training molecules, calculated with 2048-bit ECFP4 fingerprints (Morgan radius 2). The compound structures and SMILES are listed in Tables VS1 and VS2. No designed compound has an exact training match or a scaffold match.

Code	Exact training matches	Training molecules with same scaffold	Highest ECFP4 Tanimoto similarity
<i>Series I</i>			
Ia	0	0	0.4032
Ib	0	0	0.3750
Ic	0	0	0.3971
Id	0	0	0.4032
Ie	0	0	0.3750
If	0	0	0.3971
<i>Series II</i>			
IIa	0	0	0.3710
IIb	0	0	0.3500
IIc	0	0	0.4091
IId	0	0	0.3710
IIe	0	0	0.3500
IIIf	0	0	0.4091
<i>Series III</i>			
IIIa	0	0	0.3881
IIIb	0	0	0.3818
IIIc	0	0	0.4500
IIId	0	0	0.3676
IIIe	0	0	0.3818
IIIIf	0	0	0.4500
<i>Series IV</i>			
IVa	0	0	0.3714
IVb	0	0	0.3623
IVc	0	0	0.4219
IVd	0	0	0.3571
IVe	0	0	0.3559
IVf	0	0	0.4219
<i>Series V</i>			
Va	0	0	0.3662
Vb	0	0	0.3571
Vc	0	0	0.3731
Vd	0	0	0.3521
Ve	0	0	0.3380
Vf	0	0	0.3731
<i>Series VI</i>			
VIa	0	0	0.3803
VIb	0	0	0.3714
VIc	0	0	0.3881

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Table VS3 (continued)

Code	Exact training matches	Training molecules with same scaffold	Highest ECFP4 Tanimoto similarity
VIId	0	0	0.3662
VIe	0	0	0.3521
VIIf	0	0	0.3881
<i>Series VII</i>			
VIIa	0	0	0.2892
VIIb	0	0	0.2963
VIIc	0	0	0.3333
VIIId	0	0	0.2927
VIIe	0	0	0.2963
VIIIf	0	0	0.3333
<i>Series VIII</i>			
VIIIa	0	0	0.2911
VIIIb	0	0	0.2821
VIIIc	0	0	0.3108
VIIId	0	0	0.2892
VIIIe	0	0	0.2805
VIIIIf	0	0	0.3108
<i>Series IX</i>			
IXa	0	0	0.4627
IXb	0	0	0.4348
IXc	0	0	0.4265
IXd	0	0	0.4412
IXe	0	0	0.4143
IXf	0	0	0.4058
<i>Series X</i>			
Xa	0	0	0.3714
Xb	0	0	0.3582
Xc	0	0	0.3571
Xd	0	0	0.3429
Xe	0	0	0.3506
Xf	0	0	0.3623